Title: *Open Source Audit Report*

Part A1 — Origin of VLC Media Player

The World Before the Software

Before learning about VLC, I didn't really think how difficult it was to play videos earlier. Now everything works easily, but before that it was not same. Different video files needed different codecs, and if that codec was not installed, the video will not play. Many time peoples had to go on internet and search for codecs, download them, and install, which was confusing for normal users.

Also, streaming video from one system to another was not easy at all. There was very less software available for this, and most of them was either paid or not accessible to students. So overall, watching or sharing videos was not simple like today.

The Creators and the Problem

VLC was created by students of Ecole Centrale Paris in late 1990s. What I found interesting is that they were not trying to make some big software in starting. They just had a problem in their university.

They wanted to stream videos over their campus network, but the network was basic and there was no free tool which can do this work properly. Instead of using expensive software, they decided to create their own program. This project later became VideoLAN.

Why They Shared It

At first, this software was only used inside the university. But later they understood that many other people also facing same problem. So in 2001, they made it open-source and released it for everyone.

I think this was a very important decision. Instead of keeping it private or selling it, they allowed anyone to use and improve it. Other developers started adding support for more formats and fixing problems.

Because of this, VLC slowly became very powerful. Today it can play almost any video without needing extra codecs, which was the main problem before. So in a way, it solved the exact issue it was created for.

Part A2: The License (GPL)

1. The Four Freedoms

From what I understood, free software is based on four main freedoms. VLC uses GPL license, and it gives all these freedoms.

Freedom 1: You can run the software for any purpose. There is no restriction on how you use it.

Freedom 2: You can see the source code and understand how it works. If you want, you can also change it according to your need.

Freedom 3: You can copy the software and share it with other people. Like you can give it to your friends also.

Freedom 4: If you make any changes or improvements, you can share that new version also with others.

So overall, VLC is not just free to use, but also free to learn and modify.

2. Can a company modify it and sell it without sharing?

No, they cannot fully keep it private. This is because VLC uses GPL license which has something called “copyleft”.

From what I understood, this means if a company takes VLC and changes it and then sells or gives it to people, they must also share the source code of their modified version. They cannot hide it.

So they can sell it, but they cannot make it closed-source. This rule makes sure that software always stays open.

3. GPL vs MIT License

GPL and MIT are both open-source licenses, but they are very different.

GPL is more strict. If someone uses GPL code and shares their software, then they also have to make their code open. So it forces sharing.

MIT is more free in a different way. It allows people to do almost anything with the code. A company can take MIT code, modify it, and even keep it private and sell it.

So I feel GPL focuses more on keeping things open, while MIT gives more freedom to developers and companies.

4. “Free Beer” vs “Freedom”

This part was little confusing at first, but I understood it like this:

“Free beer” means you don’t have to pay money. It is just about price.

“Freedom” means you have control over the software. You can study it, change it, and share it.

In case of VLC, it is both. It is free to download, so no money is needed. But also, it gives full freedom because its source code is open.

For example, if someone knows programming, they can take VLC code and add new features in it.

5. Which license I would choose

If I build my own software, I think I would choose GPL. The reason is that I would want my work to stay open and not become closed by some company.

I feel if people are using my code, they should also share their improvements. This way everyone can benefit, just like VLC improved over time.

Part A3: The Ethics of Open Source

1. Should all software be open source? (For and Against)

From what I understand, there is no one correct answer for this. There are good points on both sides.

For open source:
I feel open source is very helpful for society. When software is open, anyone can learn from it, especially students. Bugs can also be fixed faster because many people are working on it. Also, people who cannot afford expensive software can still use good tools for free. For example, if VLC was not open-source, many people may not have access to such a powerful media player.

Against open source:
At the same time, I also think not all software can be open. Some software takes a lot of time, money, and effort to build. For example, big companies spend huge money on games or medical software. If they make everything open-source, they may not earn profit and cannot pay their employees. So in some cases, keeping software closed is also understandable.

2. Is it ethical for big companies to make profit from free community work?

At first, this feels a bit unfair to me. If a company takes free software made by volunteers and then earns a lot of money from it without giving anything back, it does not seem right.

But when I looked more into it, I saw another side also. Some companies like **Red Hat** and **Google** actually contribute a lot to open-source projects. They hire developers and pay them to improve these tools.

So in my opinion, it is ethical only if companies give something back. This can be in form of code, support, or improvements. If they only take and don’t contribute, then it feels wrong.

3. “Standing on the shoulders of giants” and innovation

This phrase was new to me, but I understood it in a simple way. It means we don’t have to start everything from zero. We can use the work already done by others and build something better on top of it.

In open source, this happens a lot. For example, developers can use existing tools like VLC and build new applications instead of creating everything from beginning.

I think this actually helps innovation. Because developers save time on basic things, they can focus more on new ideas. So instead of slowing down progress, open source makes it faster.

Part B: The Linux Footprint of VLC

1. Installation Method

When I installed VLC on my Linux system, I noticed that it was very easy and did not require any complex steps. I did not need to download source code or compile anything manually.

I just used the terminal and typed:

```
sudo apt update
sudo apt install vlc
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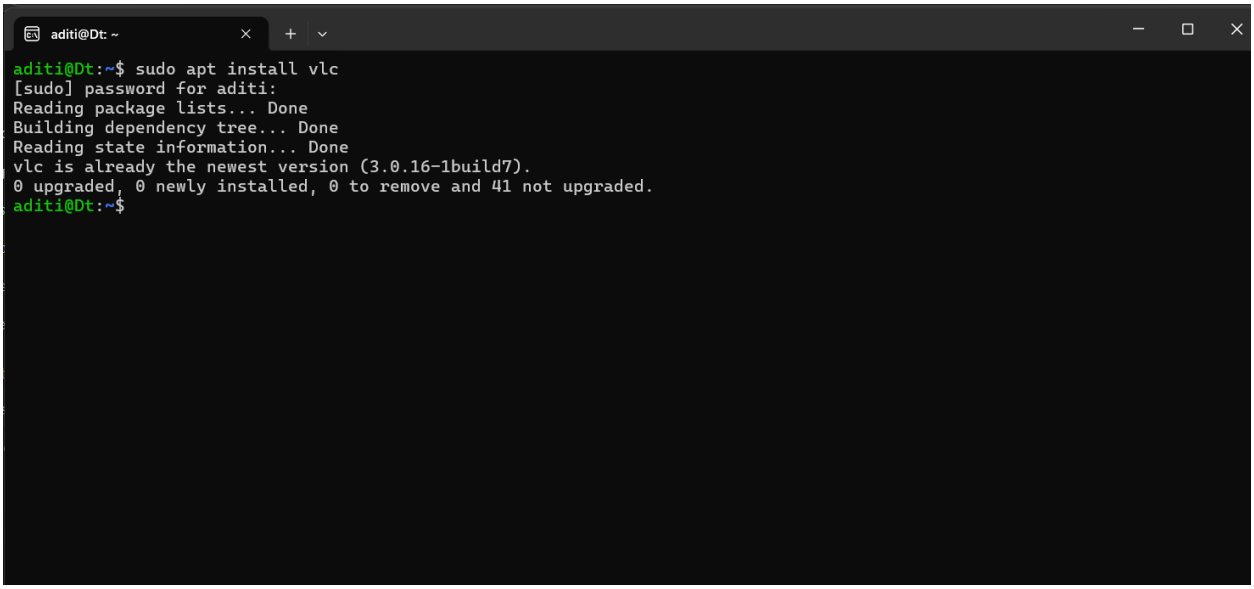
From what I understand, VLC is available through the apt package manager in Ubuntu-based systems. This makes installation fast and also safe, because the software comes from official repositories which are verified.



2. Key Directories

After installing VLC, I explored where its files are stored in the system. I used the command:

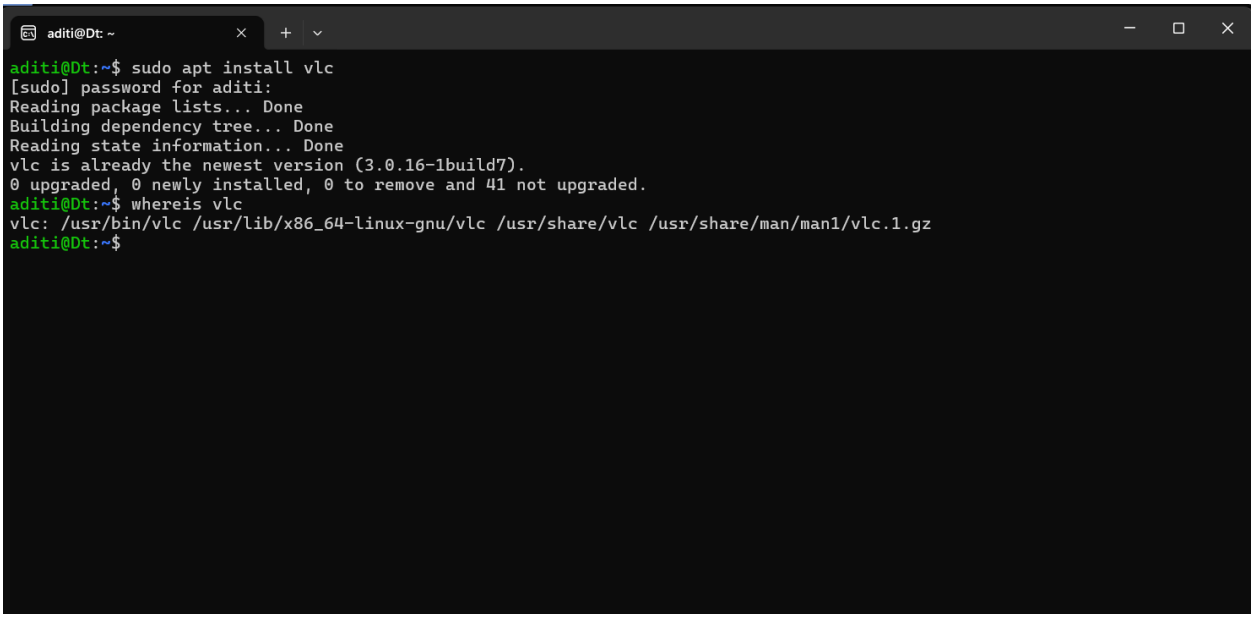
whereis vlc

This helped me see different locations related to VLC.

The main executable file is located in /usr/bin/vlc, which is the actual program that runs when we open VLC.

For configuration, I found that VLC does not store settings in system folders like /etc. Instead, it keeps user-specific settings inside the home directory at ~/.config/vlc. This means each user can have their own settings.

There are also library files stored in /usr/lib, which are used internally by VLC to handle media playback.



3. User and Group (Security Focus)

One interesting thing I noticed is that VLC runs as the normal logged-in user, not as root.

When I tried to run:

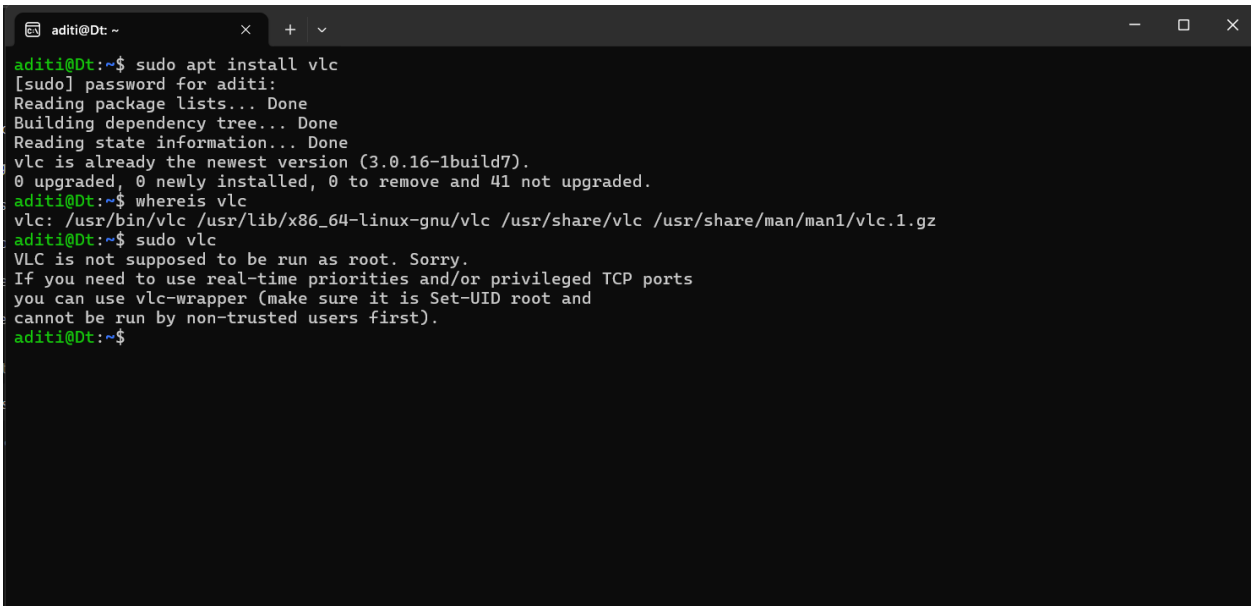
sudo vlc

it actually gave an error and refused to start.

At first I thought something is wrong, but later I understood that this is done for security reasons.

Media files can sometimes contain harmful code. If VLC runs as root and something goes wrong, it can affect the whole system. But if it runs as a normal user, the damage is limited.

So I feel this is a smart design choice for safety.



4. Service Management

VLC is not like a background service such as a web server. It is mainly a desktop application.

So we don’t use commands like:

systemctl start vlc

Instead, we simply run it by typing:

vlc

or by clicking the icon.

To stop it, we just close the application or press Ctrl+C in the terminal.

I also found that there is a command called cvlc, which runs VLC in terminal mode without a graphical interface. This can be used for advanced purposes like streaming.

5. Update Model

From what I understood, updates for VLC come through the Linux package system.

When developers from the VideoLAN community fix bugs or security issues, they release updates. These updates are then handled by the maintainers of the Linux distribution.

On my system, I can get these updates by running:

sudo apt upgrade

So basically, the community creates the fix, the Linux distribution checks it, and then it is delivered to users through the package manager.

Part C: The FOSS Ecosystem of VLC

1. Dependencies (What VLC depends on)

From what I found, VLC does not work completely alone. It depends on many other open-source libraries to function properly.

One important dependency is **FFmpeg**, which helps VLC to decode and encode different audio and video formats. Without this, VLC would not be able to support so many file types.

Apart from this, VLC also uses different codecs and system libraries provided by Linux. So it is clear that VLC is part of a bigger ecosystem and not a single independent software.

2. What VLC has enabled or inspired

I think VLC has made a big impact in how people use media players today. Before VLC, users had to install codecs separately, but VLC made it simple by supporting almost all formats in one place.

Because of this, many other media players and tools have been inspired to provide similar “all-in-one” functionality.

Also, VLC is used in many areas like streaming, video conversion, and even broadcasting. So it has influenced not just users but also developers.

3. Relation to Web / LAMP Stack

VLC is not directly part of the LAMP stack (Linux, Apache, MySQL, PHP), because it is mainly a media player and not a web server or database.

However, it can still be used along with web technologies. For example, VLC can stream video content, which can then be accessed through web applications.

So even though it is not a core part of the web stack, it still connects to it in some ways.

4. Community and Governance

VLC is developed by the **VideoLAN** community.

From what I understood, this is a group of developers from different parts of the world who contribute to the project. They communicate through forums, mailing lists, and online platforms.

There is no single company controlling VLC. Instead, it is managed by the community, which is one of the main ideas of open-source software.

Part D: Open Source vs Proprietary Software

Comparison: VLC vs Windows Media Player

Feature	**VLC (Open Source)**	**Windows Media Player (Proprietary)**
Cost	Free to download and use	Comes with Windows (not fully free separately)
Security	Code is open, so anyone can check for bugs	Code is closed, only company can check
Support & Reliability	Community support, regular updates	Official support from company
Freedom to Modify	Anyone can modify and share	No modification allowed
Control	Community-driven	Controlled by company
Overall Verdict	More flexible, free, and powerful for most users	Limited flexibility but stable for basic use

From this comparison, I feel that both open-source and proprietary software have their own advantages.

VLC is very flexible and powerful. It supports many formats and gives full freedom to users. Also, since it is open-source, anyone can improve it. I personally found VLC more useful because it works without needing extra setup and is completely free.

On the other hand, proprietary software like Windows Media Player is more controlled and may provide better official support. But it does not give freedom to modify or understand how it works.

Overall, I would recommend VLC for real-world use, especially for normal users and students. It is simple, reliable, and works in most situations.

If I get a chance, I would like to contribute to open-source projects like VLC in the future. Even small contributions like fixing bugs or improving features can help the community.